



Web programming I

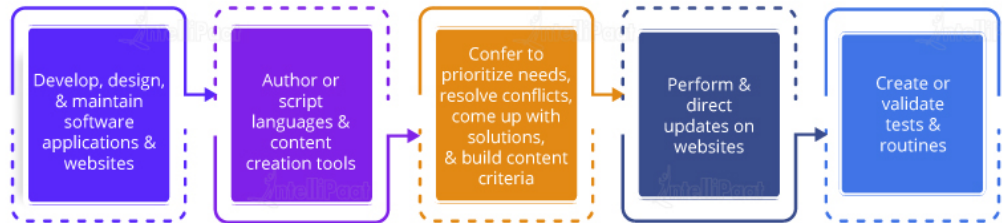
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Tasks of a web developer

Major Roles and Responsibilities of a Web Developer



Code the client-side of a web application.

Converts the design into a application using HTML, CSS and JavaScript.

Ensure responsive and adaptive User Interface



Frontend Developers	Backend Developers	Full Stack Developers
Code the client-side of a web application	Build and maintain the services needed by a web application behind the scenes	Experts in both frontend and backend development
Converts the design into a real-life application using HTML, CSS, and JavaScript	Ensure that the web applications keeps running smoothly	Supervise the development and guide on best practices
Ensure responsive and adaptive User Interface	Manage services like API, data storage for the web application	Know both business requirements and user interface requirements

WHAT DOES A WEB DEVELOPER DO?

FRONTEND DEVELOPERS

- Code the frontend of a website; i.e. the part that the user sees and interacts with
- Bring the web designer's designs to life using HTML, JavaScript and CSS
- Ensure responsive design

BACKEND DEVELOPERS

- Work behind-the-scenes, building and maintaining the technology needed to power the frontend
- Ensure that everything the frontend developer builds is fully functional
- Create and manage the database

FULL STACK DEVELOPERS

- Experts in both backend and frontend development
- Guide on strategy and best practices
- Well-versed in both business logic and user experience



A Day in the Life of a Web Developer

Build user interfaces and prototypes from wireframes and designs



Code and deploy applications in a cross-platform, cross-browser environment



Build products using HTML/CSS/JS

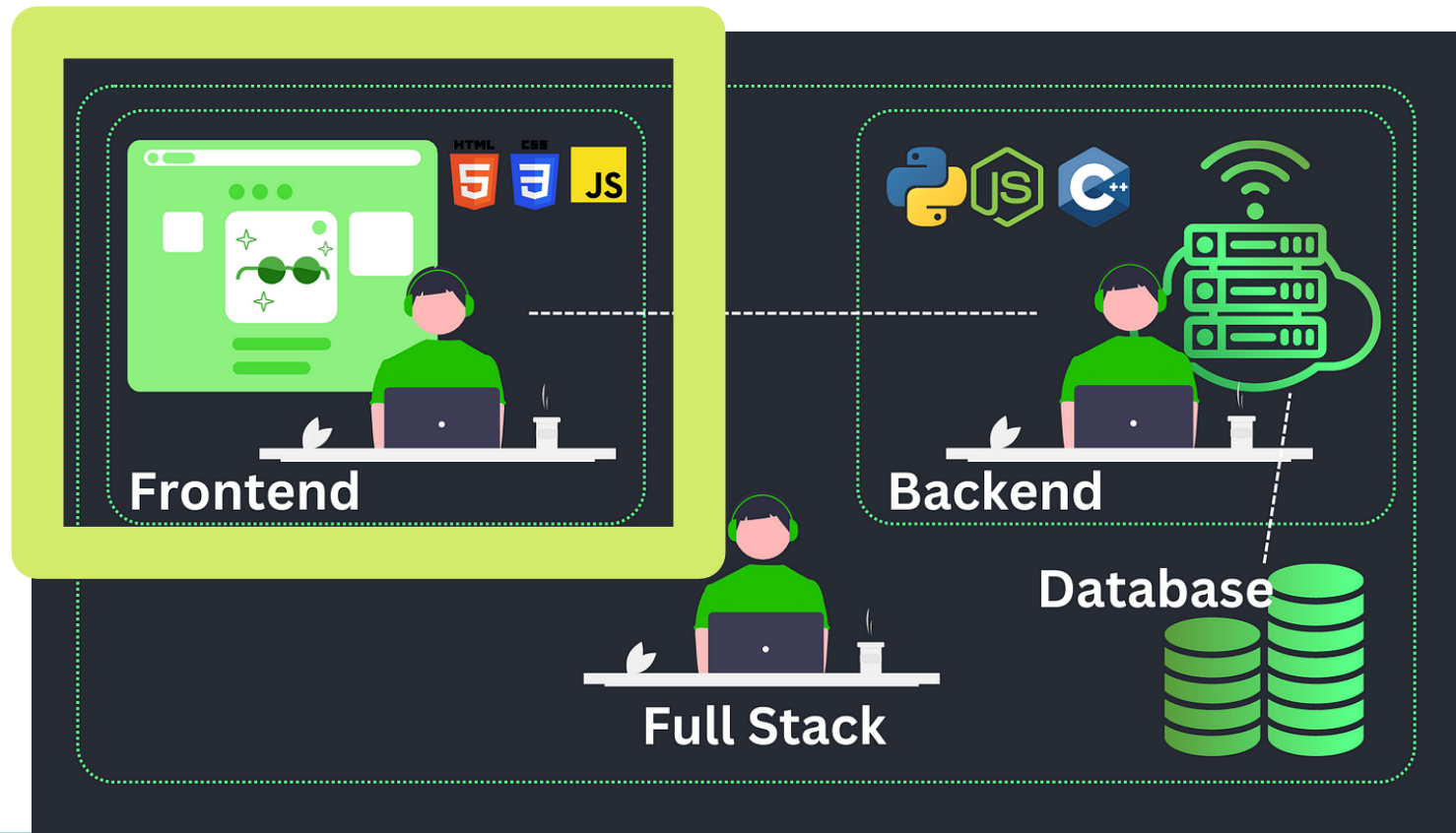


Update current websites to meet modern web standards



Course introduction > Focus of the course

Focus of the course





Course introduction > Course technologies

Course technologies





Course introduction > Course entry poll

Course entry poll

Go to

www.menti.com

Enter the code

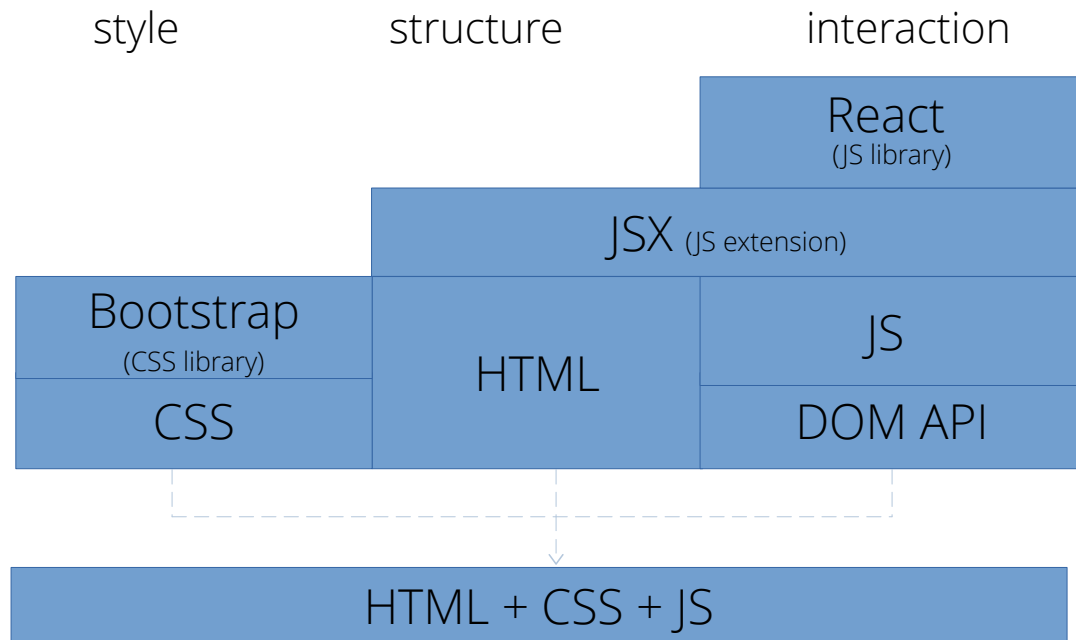
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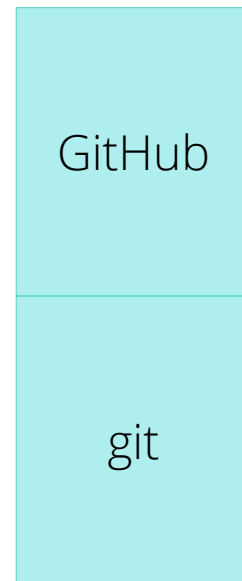
Or use QR code



Course technologies



collaboration
version control

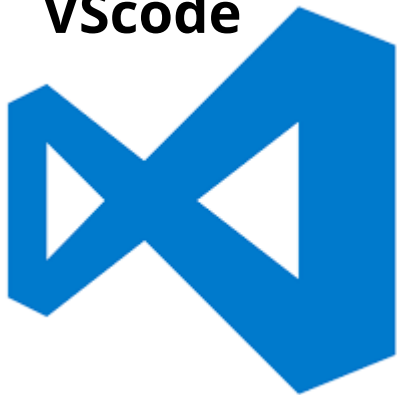




Course introduction > Software to install

Software to install

VScode



<https://code.visualstudio.com/>



git

<https://git-scm.com/>



Course platform on UEF Moodle

<https://moodle.uef.fi/course/view.php?id=15158>

Join using the enrolment key corresponding to your exercise group.

Exercise group	Enrolment key
Joensuu, Finnish R1	WOHI-JNS-FI-R1
Joensuu, Finnish R2	WOHI-JNS-FI-R2
Joensuu, English R1	WOHI-JNS-EN
Kuopio, Finnish R1	WOHI-KUO-FI-R1
Kuopio, Finnish R2	WOHI-KUO-FI-R2
Kuopio, English R1	WOHI-KUO-EN



Course objectives

On study guide	Student is able to
Understand the client-server architecture and HTTP.	identify and explain the principles of client–server architecture and the HTTP protocol.
Use HTML, CSS, JavaScript, and Bootstrap for building interactive and responsive web pages.	design and implement interactive and responsive web pages using HTML, CSS, JavaScript, and Bootstrap.
Apply version control systems (Git) and collaborative workflows (GitHub) in web projects	use version control with Git and collaborative workflows with GitHub in web development projects.
Analyse user-interface requirements to design efficient, reusable, and component-based web applications with React.js.	analyze user interface requirements and design reusable, component-based web applications using React.
Integrate external APIs into web application using asynchronous JavaScript.	integrate external APIs into a web application using asynchronous JavaScript.
GENERIC SKILLS	GENERIC SKILLS
Problem-solving	solve web development problems using appropriate tools and methods
Critical thinking	evaluate alternative design solutions using critical thinking
Creativity	design original web solutions demonstrating creativity



Course assessment

Item	Points
Final project (due 13.11.2026) **	40%
AI assignment (due 13.11.2026)**	20%
Exercises (from weekly sessions) *	10%
Final exam 23.10.2026**	30%

* Attendance is not mandatory but submitting the exercises done in class will be 10% of the grade

** Passing these assessments independently is mandatory for passing the course



Course content

Course topics

- web basics: Intro to the web, HTML, CSS, Accessibility, and Bootstrap
- version control
- JavaScript (basics, libraries, REST APIs)
- React (components, state, routing, API integration)
- programming with AI
- course review

Key dates

course exam	23.10.2026
final work deadline	13.11.2026
exam retake	27.11.2026



Final project – overview

Create a **web application** of a topic that interests you.

The application must

- be developed using **React**
- have at least **3 different "screens"** with different features
- use **third-party libraries** for both style and functionality
- use an **external API** (e.g. from <https://github.com/public-apis/public-apis>)



Final project – some ideas

- Recipe finder
- Crypto price tracker
- Travel planner
- Quiz game
- Custom chatbot with LLM
- Game puzzle

- No weather apps!

Try to do something that is useful/interesting to you!